

NIMBA TOTEM LIGHT

Smart Solar Community Beacon

Flood / Heavy Rain Alert System



NIMBA SYMBOLIC CROWN

Cultural identity integrated around the solar panel

360° LED-LIGHT RING

Several small DC LED modules facing outward inside the enclosure for even community illumination

WEATHER STATION

Rain / flood sensor (precipitation sensor) Detects heavy rain and flooding risk

INTERNAL WIRING

All cables protected inside the pole for safety and durability

USB CHARGING PORT WITH 30 MIN TIMER

Mechanical timer controls power to the USB port. Automatically cuts off after 30 minutes.

BATTERY & CONTROL ENCLOSURE (BASE)

LiFePO₄ Battery
MPPT Controller
All electronics protected inside a waterproof enclosure

TOP VIEW (SOLAR PANEL) WITH NIMBA CROWN



KEY FEATURES

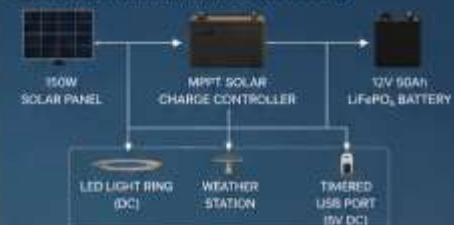
- 100% OFF-GRID
- 360° COMMUNITY LIGHTING
- FLOOD / HEAVY RAIN ALERT
- NIMBA CULTURAL IDENTITY
- DURABLE & WATER RESISTANT
- USB CHARGING PORT (30 MIN AUTOMATIC TIMER)

UNDER PANEL VIEW (ENCLOSURE)



LED MODULES
Positioned around the inside perimeter facing outward

ELECTRICAL DIAGRAM (INSIDE ENCLOSURE)



USB PORT TIMER – MECHANICAL (NO ENERGY USE)



- Maximum 30 minutes per turn
- No electricity needed
- Automatically cuts off power
- Simple, reliable, and durable

SUGGESTED SPECIFICATIONS

- Solar Panel: 150W Monocrystalline
- Battery: 12V 50Ah LiFePO₄ (expandable)
- LED Lighting: 30W - 60W DC LED modules
- LED Color Temp: 3500K Warm White
- Weather Sensor: Rain / Precipitation Sensor
- Autonomy: 2 - 3 days (depending on usage)
- Waterproof Rating: IP65 - IP66
- Structure: Galvanized Steel / Aluminum
- Mount Height: 4m - 5m (adjustable)

USB TIMED ACCESS – HOW IT WORKS

1. CONNECT

User connects phone or device to the USB port.



2. SET TIMER

User turns the mechanical timer knob up to 30 minutes.



3. POWER ON

Power is supplied to the USB port for the selected time.



4. AUTO OFF

After 30 minutes, power cuts off automatically. Port is disabled until next turn.



FAIR COMMUNITY ACCESS



Encourages sharing and equal access



Prevents overuse and long occupation



No energy waste



Promotes responsibility and community spirit

NIGHT VIEW – NORMAL MODE



360° LIGHT
Warm white lights up the community



FLOOD ALERT
Blue flashing light warns the community



OFF-GRID
Solar powered fully independent infrastructure



A CULTURALLY INSPIRED, OFF-GRID SMART LIGHTING SOLUTION DESIGNED FOR RESILIENCE, COMMUNITY SAFETY, AND ENVIRONMENTAL AWARENESS.



CULTURE IDENTITY



TECHNOLOGY INNOVATION



COMMUNITY EMPOWERMENT

NIMBA TOTEM LIGHT – END TO END ARCHITECTURE DESIGN

Smart Solar Community Beacon – Flood / Heavy Rain Alert System (100% Offgrid)



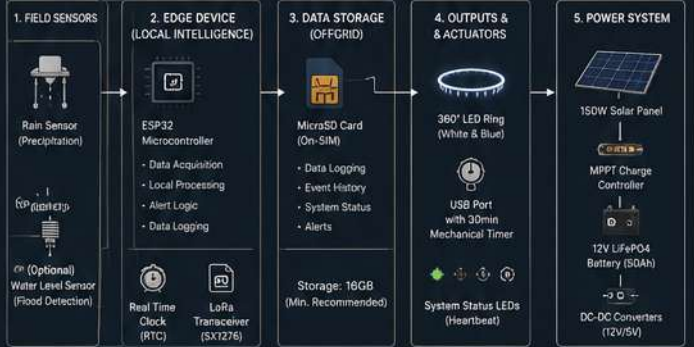
KEY FEATURES

- 100% OFFGRID
- 360° COMMUNITY LIGHT
- FLOOD / HEAVY RAIN ALERT
- NIMBA CULTURAL IDENTITY
- USB CHARGING PORT WITH 30 MIN TIMER
- DATA STORED LOCALLY
- DURABLE & WATER
- LOW POWER, HIGH EFFICIENCY

RAIN / FLOOD ALERT BEHAVIOR

When heavy rain is detected, the light ring turns BLUE and flashes to alert the community.

END-TO-END SYSTEM ARCHITECTURE



NETWORK / COMMUNICATION OVERVIEW



HARDWARE ARCHITECTURE & SPECIFICATIONS

A. POWER SYSTEM

COMPONENT	SPECIFICATION
Solar Panel	150W Monocrystalline, 18V, 8.3A
MPPT Charge Controller	12V/24V Auto, 20A
Battery	12V 50Ah LiFePO4
DC-DC Converter 1	12V to 5V (For ESP32, Sensors)
DC-DC Converter 2	12V to 12V (LED Ring)
Fuse Protection	Inline Fuses (Per Circuit)
Surge Protection	TVS Diode (12V System)

B. LIGHTING SYSTEM

LED Module Type	SPECIFICATION
Module	DC LED Modules
Voltage	12V DC
Power	30W (White) + 10W (Blue Alert)
Configuration	360° Ring (Under Panel)
Lumen Output	~3500 Lumens (White Mode)
CCT	3500K Warm White
Protection	IP65 Waterproof

C. SENSORS

COMPONENT	SPECIFICATION
Rain Sensor	Tipping Bucket Rain Gauge (0.2mm)
Flood Sensor (Optional)	Water Level Sensor (Ultrasonic or Float)
Temperature (Optional)	DS18B20 (Waterproof)
Humidity (Optional)	DHT22 (Waterproof)

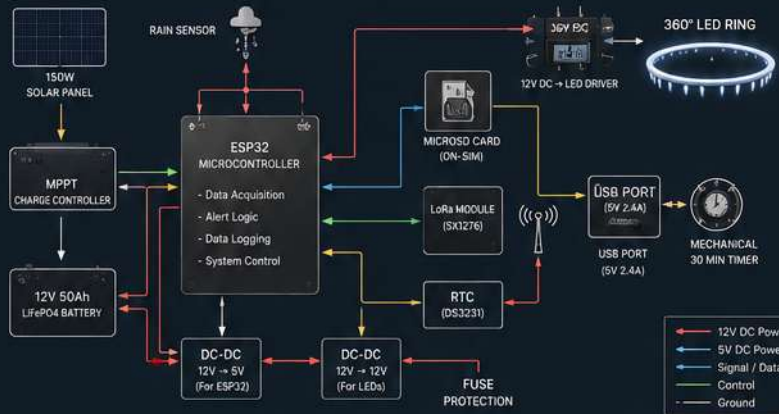
D. EDGE DEVICE & COMMUNICATION

COMPONENT	SPECIFICATION
Microcontroller	ESP32 (Low Power)
LoRa Module (Optional)	SX1276 (868/915MHz)
RTC	DS3231 (Accurate Timekeeping)
Storage	MicroSD Card (On-SIM) 16GB Min.
Indicators	Status LEDs (Power, Alert, System)

E. USB CHARGING WITH TIMER

USB Port	Dual USB 5V/2.4A Waterproof
Timer	Mechanical Timer (30min)
Mechanism	Countdown → Auto power cut-off
Reset	Manual Reset After 30min

SYSTEM BLOCK DIAGRAM



DATA LOGGING & STORAGE (OFFGRID)



EXAMPLE FILE STRUCTURE



ALERT LOGIC FLOW



USB TIMER MECHANISM (MECHANICAL)



COMMUNICATION (OPTIONAL – LORAWAN)



MECHANICAL & ENVIRONMENTAL

Enclosure Protection	IP66
Material	Galvanized Steel / Aluminum
Mount Height	4 - 5 m (Adjustable)
Wind Resistance	Up to 120 km/h
Operating Temp.	-20°C to 60°C
Humidity	0 - 100% RH
Lifespan	10+ Years (Design Target)

SOFTWARE (EDGE)

- FreeRTOS (ESP32)
- C/C++
- Sensor Libraries
- SD Logging Library
- LoRa Library (Optional)
- RTC Time Sync
- Watchdog for Reliability

POWER CONSUMPTION ESTIMATE

COMPONENT	POWER (W)	HOURS/DAY	DAILY (WH)
LED Ring (White Avg)	20W	6h	120Wh
ESP32 + Sensors	1W	24h	24Wh
LoRa (Avg)	0.2W	24h	4.8Wh
USB Port (Occasional)	10W	0.5h	5Wh
TOTAL	~30W		~154Wh

ENERGY BALANCE

Solar Panel: 150W
Peak Sun Hours: 5
Daily Generation: ~750Wh
Daily Consumption: ~154Wh
Surplus: ~596Wh
→ System is energy positive